



GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/ INFORMATION SYSTEMS

FUNDAMENTALS OF DATABASE MANAGEMENT SYSTEMS

TUTORIAL 01

Instructions:

Answer **ALL** questions

Duration: 03 Hours

Question 01

[30 Marks]

a) Consider the given scenario and answer the following questions.

Assume a CONFERENCE_REVIEW database in which researchers submit their research papers for consideration. Reviews by reviewers are recorded for use in the paper selection process. The database system caters primarily to reviewers who record answers to evaluation questions for each paper they review and make recommendations regarding whether to accept or reject the paper. The data requirements are summarized as follows:

- Authors of papers are uniquely identified by e-mail id. First and last names are also recorded.
 - Each paper is assigned a unique identifier which is generated by the system (Paper_ID). Paper is identified by a title, abstract, and the name of the electronic file containing the paper.
 - System need to record all the details of the authors. A paper may have multiple authors and one of the authors among them is designated as the contact author.
 - Reviewers of papers are uniquely identified by e-mail address. Each reviewer's first name, last name, phone number, affiliation, and topics of interest are also recorded.
 - Each paper is assigned between two to four reviewers. The assigned reviewers rate each paper on a scale of 1 to 10 in four categories: technical merit, readability, originality, and relevance to the conference.
 - Finally, each reviewer provides an overall recommendation regarding each paper. Each review contains two types of written comments: one to be seen by the review committee only and the other as feedback to the author(s).
- I. Examine the given scenario and design an entity–relationship diagram for the CONFERENCE_REVIEW database.

(15 Marks)

- II. Formulate the DML statements to create at least two relational tables for the CONFERENCE_REVIEW database. (Use CONSTRAINTS to reference foreign keys to primary keys and appropriate data types).
(4 Marks)
- III. Write DML statements to insert at least two records for the created tables in (II).
(4 Marks)
- IV. Assume that, once you created the tables and inserted records, you are going to delete a record from one of the tables you created. But you are not allowed to perform the operation as violation of the referential integrity rule.
- 1) Identify possible operation to overcome this and delete the record from table.
(3 Mark)
 - 2) Compose your solution by writing the relevant statements.
(4 Marks)

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Question 02

[20 Marks]

Considering the Figure 1 ER diagram of a Bank, map the ER Diagram to a relational schema. (Define attributes if necessary).

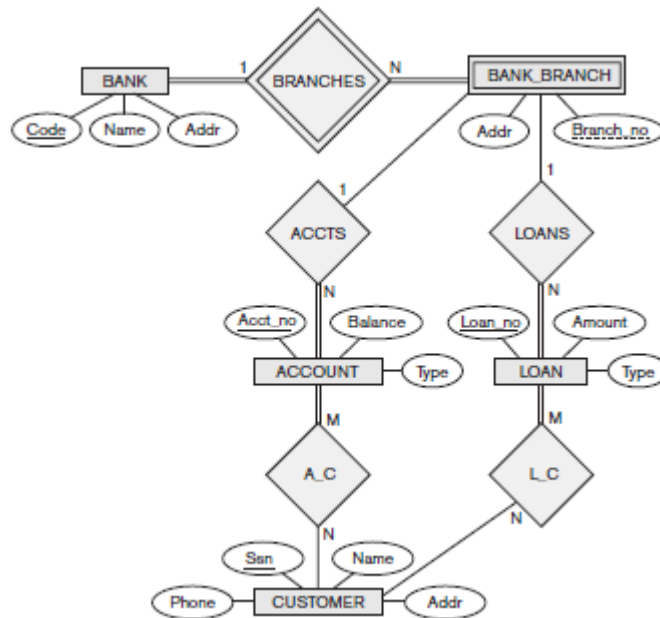


Figure 1 Bank

Database

Question 03

[25 Marks]

Consider the given relational schema of a hotel database.

Hotel (HotelNo, HotelName, City)

Room (RoomNo, HotelNo, Type, Price_Per_Night)

Booking (HotelNo, GuestNo, DateFrom, DateTo, RoomNo)

Guest (GuestNo, GuestName, GuestAddress)

a) Formulate SQL statements to express the given queries.

I. Display Names of the of all hotels located in London.

(2 Marks)

II. List the names and addresses of all guests currently accommodate in London, alphabetically ordered by name.

(3 Marks)

III. How many hotels are available in the database? retrieve and display by column name "NO OF HOTELS"

(5 Marks)

IV. Retrieve the names of the hotels which has double or family rooms with a price below £40.00 per night,

(5 Marks)

V. Due to seasonal conditions, Higher Management of the hotel “*GLANCE*” decided to increase the price of a double room by £10.00 per night. Formulate DML statement to make necessary changes of the records.

(5 Marks)

b) Formulate relational algebra expressions for the following operations.

I. Display Names of the of all hotels located in “Salisbury”.

(2 Marks)

II. List the names and addresses of all guests currently living in “Portsmouth”.

(3 Marks)

Question 04

[25 Marks]

a) Briefly discuss the possible ways of identifying the following.

I. A relation is in Unnormalized form

II. A relation is in 1NF

III. A relation is in 2 NF

IV. A relation is in 3 NF

V. Transitive Dependencies

(5 x 2 Marks)

b) Analyze the given records of Employee_Dep relational table and answer the following questions.

Table 1 Employee_Dep relation

EmpNo	DepNo	Emp_Name	EmpAddress	DepName	Salary_Slip_No	Salary
E001	DP1	Rose	London	CS	1551	£750
E002	DP2	Ann	Bath	CE	1564	£800
E003	DP2	Christine	Cambridge	CE	1434	£600
E004	DP3	Sheril	Canterbury	IT	1412	£500

Employee_Dep table Primary Key : (EmpNo, DepNo)

EmpNo → Emp_Name, EmpAddress, DepNo

DepNo → DepName

Salary_Slip_No, → Salary

EmpNo, DepNo → Salary_Slip_No, Salary

I. In which Normal Form the relation is in?

(1 Mark)

II. Decompose the relation into the highest Normal Form (NF).

(4 Marks)

c) Refer the following relational tables and answer the questions.

Employee Table

	EmpId	EmpName	EmpAddress	salary
1	E001	Bob	Amstersam	25000.90
2	E002	Jan	Austria	50000.00
3	E003	Martin	Mexico	75000.78
4	E004	Shroff	India	85000.90
5	E005	Mick	Stafford	100000.90

DepNo	DepName	EmpNo
D1	Soft Eng	E001
D2	Comp Science	E002
D3	Comp Eng	NULL
D4	Infor Tec	NULL
D5	Com Sce	NULL

Instruction: Always use the Employee table as left table when formulate queries.

side

I. Write the SQL Queries to generate following results

1)

EmpName	salary	DepName
Bob	25000.90	Soft Eng
Jan	50000.00	Comp Science
Martin	75000.78	NULL
Shroff	85000.90	NULL
Mick	100000.90	NULL

2)

EmpName	salary	DepName
Bob	25000.90	Soft Eng
Jan	50000.00	Comp Science
NULL	NULL	Comp Eng
NULL	NULL	Infor Tec
NULL	NULL	Com Sce

(2x 5Marks)

---- END OF THE PAPER ----